Talk in binary

IoT · Level 2 · Lesson 3



Today you will

- Count from 0 to 31 in binary with place values.
- Explain that every reading, message and photo is stored as bits.
- Work out why more bits mean more detail but more data.

Bits and bytes

- Each place is worth double
- Read from the left
- 8 bits = 1 byte: 0 to 255
- More bits: more detail, more data

Binary cards

1. Make five dot cards
2. Make your age
3. Make two more numbers
4. Count to 31 on one hand
5. Check your name's bytes

No kit needed. When you count on your hand, keep your palm facing you, so no finger pattern looks rude to anyone watching.

Let’s discuss

1. Why do computers use 1 and 0 instead of the ten digits we use?
2. Why does the same message take more data in Arabic than in English?
3. When is it worth paying for more bits, and when isn't it?

Check yourself

- Write 22 in binary, and say how many values 6 bits can count.